Document 08 — System Diagram and Outsourcing Map

Operator	[COMPANY NAME], trading as PlayStake
Owner	[COMPLIANCE LEAD NAME], Head of Compliance (with engineering)
Version	0.1 — draft for licence application
Review	On any new supplier, hosting change or new product; at least annually
Relates to	Remote operating licence application: system diagram with commentary; LCCP 15.2.1 key events (outsourcing of key functions); RTS security requirements (Document 13)

> Draft prepared from the platform as built on 30 September 2026. Items in `[BRACKETS]` must be completed before submission. Where a supplier is "planned", no contract exists yet and the function is not live.

1. Purpose

This document gives the Commission a single picture of the PlayStake gambling system: every component, who provides and operates it, where it runs, and how data and money flow between components. It also lists every function we rely on a third party for (the outsourcing map). Document 09 follows a customer through the system end to end. Document 10 maps each activity to its location, provider and operator.

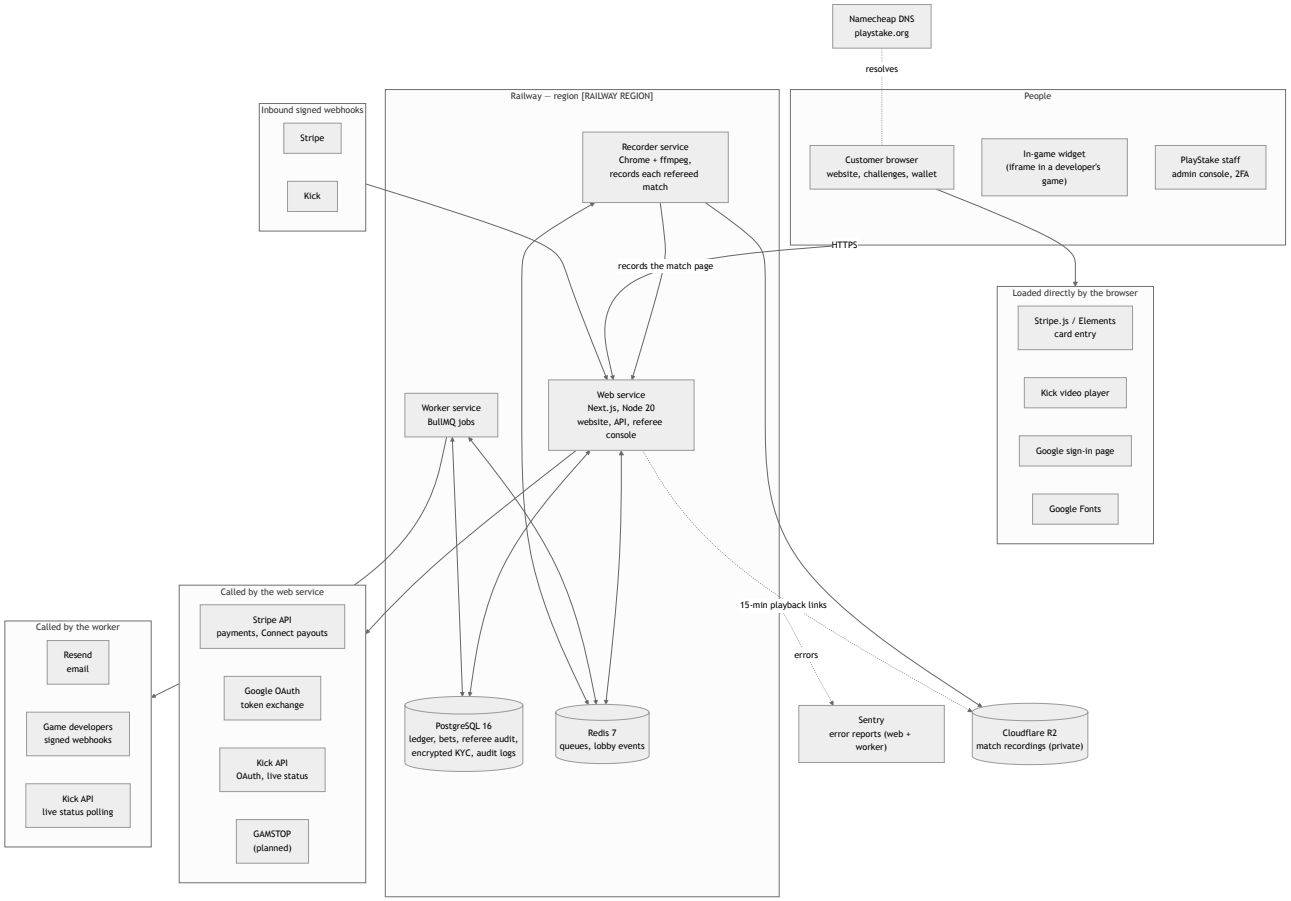
2. Summary

PlayStake is a single web application with a separate background-worker process, a PostgreSQL database and a Redis instance. All four are hosted by one provider (Railway). PlayStake writes and operates all of its gambling software itself: challenges and match management, the referee console, the wallet and ledger, result handling, and all compliance controls. No third-party gambling software or platform is used. The games themselves (EA SPORTS FC 27 and Tekken 8) are third-party console and PC games that players run on their own devices; PlayStake does not host, run or modify them, and uses no random number generator.

A third service, the **recorder**, records every refereed match: it opens a private PlayStake page showing both players' and the referee's Kick streams, captures it with sound, and stores the sealed video in Cloudflare R2 (Document 18).

Third parties provide: card payments and payouts (Stripe), match-recording storage (Cloudflare R2), transactional email (Resend), error monitoring (Sentry), optional sign-in (Google), live-stream integration (Kick), domain name registration and DNS (Namecheap), source control and build checks (GitHub). The national self-exclusion check (GAMSTOP) and dispute resolution (IBAS) are contracted but not yet live.

3. System diagram



3.1 Commentary

- **The recorder.** A third service, from the same code, takes a recording job from Redis when a referee starts a match, records the private match page in Chrome with ffmpeg, uploads 5-minute pieces to R2 while recording, and seals the final file into the audit trail when the result is recorded (Document 18).
- **One application, two processes.** The web service answers every request from customers, staff and game developers. The worker service runs everything time-based: settlement after the dispute window, expiry of unanswered bets, no-show voiding, dispute escalation, anomaly and harm detection, AML scans, the daily ledger audit, email delivery, outbound webhooks, deposit-limit activation and session cleanup. Both processes are deployed from the same code and connect to the same database.
- **The database is the system of record.** Balances exist only as ledger accounts in PostgreSQL, changed only through a double-entry `transfer()` that writes two balancing entries. Ledger entries, the admin audit log, referee audit events and the historical game events of the withdrawn on-site games cannot be edited or deleted (database triggers reject it).
- **No games run on our servers.** Every match is played in a third-party game on the players' own devices and streamed on Kick. An independent human referee decides the result from both streams through the referee console. PlayStake's own on-site games and their random number generator were withdrawn on 25 October 2026. See Document 11 for testing.
- **Card data never reaches PlayStake.** Card entry uses Stripe Elements, loaded from Stripe, so PlayStake never sees or stores card numbers. Stripe tells us about payments through signed webhooks, which are verified before any balance changes.
- **Redis holds no money and no customer records.** It carries job queues, real-time lobby events and the worker heartbeat. Losing it delays background work; it does not lose funds or records.
- **Identity documents** are stored in PostgreSQL encrypted with AES-256-GCM, under a key held only in the hosting environment. Every staff view is logged.

4. Outsourcing map

"Operated by" is who runs the service day to day. "Data shared" is what leaves PlayStake. Every supplier is a key supplier in the sense of LCCP 15.2.1 unless marked otherwise; changes are notified as key events (Document 14).

Function	Supplier	Status	What they do	Data shared	Where processed	Contract / assurance
Hosting and storage	Railway Corp.	Live	Runs the web, worker and recorder services, PostgreSQL and Redis; database backups (to be confirmed as enabled, Document 13)	All platform data (hosted, not accessed by Railway for its own purposes)	Railway EU West — `europe-west4` (Amsterdam, Netherlands), all five services (checked 27 October 2026)	Railway terms and DPA. [CONFIRM SOC 2 REPORT]
Match-recording storage	Cloudflare, Inc. (R2)	Live from 27 October 2026 (DPIA sign-off outstanding, Document 18)	Stores sealed match recordings privately; deleted after 30 days, or 12 months after a dispute is decided	Video and sound of players' and referees' streams	R2 bucket with **EU jurisdiction**; no public access	Cloudflare DPA (with UK addendum). Document 18
Banking (customer funds)	[BANK NAME]	To open (Policy 19 §6)	Holds customer funds in a separate customer-funds account, used only for customer money; Stripe pays out to it; reconciled daily	Account balances and transfers	UK	Bank terms; written acknowledgement that it holds customer money [CONFIRM]
Banking (company)	[BANK NAME]	Live	Company account for PlayStake's own money (fees earned, costs)	—	UK	Bank terms
Payment services	Stripe Payments Europe / Stripe, Inc.	Live, **test mode**	Card deposits (Payment Intents), withdrawals via Stripe Connect Express, Radar fraud screening (to configure)	Customer name, email, amount; card data handled by Stripe only	Stripe (EU/US)	Stripe Services Agreement; PCI DSS Level 1; PSR authorisation and approved methods in Policy 16
Age and identity verification	Didit (didit.me)	Live (normal route)	Hosted identity check: document authenticity, liveness, face match, age (18+), proof of address, AML screening and ongoing re-screening	Name, date of birth, address, document and selfie images (biometric), email, PlayStake user id	EU (Spain); processing and storage locations to confirm	ISO 27001, SOC 2 Type II; DPA to sign; DPIA in Document 17
	In-house	Live (fallback)	Customers who can't use Didit upload an identity document; staff review it before any deposit or gambling	—	PlayStake (Railway)	—
Sanctions and PEP screening	[SCREENING PROVIDER]	Planned	Screening at verification and on list updates	Name, date of birth, nationality	[TBC]	To contract (Master Manual §8 item 5)

Function	Supplier	Status	What they do	Data shared	Where processed	Contract / assurance
Multi-operator self-exclusion	GAMSTOP (The National Online Self Exclusion Scheme Ltd)	Planned	Checks every customer against the national register	Name, date of birth, email, postcode	UK	Membership to complete (Master Manual §8 item 3)
Geolocation	None (residency only)	**Partial**	Launch territory is England, Scotland and Wales. Residency is enforced from the verified address (country GB, not a Northern Ireland BT postcode) at verification, at approval, and before every deposit and stake. There is no check of where the customer is when they play (no IP geolocation).	—	—	See §6
Fraud detection	In-house	Live	Anomaly detection (win rates, winner patterns, fast settlement, volume spikes), shared-identity checks, AML scans	—	PlayStake	—
	Stripe Radar	To configure	Card-fraud scoring on deposits	As payment services	Stripe	As payment services
Customer service	In-house, on Google Workspace email	Live	support@playstake.org (Google Workspace mailbox), complaints procedure, in-app messages	Customer emails	Google	Google Workspace terms and DPA
Alternative dispute resolution	IBAS	To sign	Independent adjudication of unresolved complaints	Complaint file, on referral only	UK	ADR agreement (Master Manual §8 item 2)
Marketing	None	—	No marketing providers, affiliates or ad networks	—	—	—
Website analytics	Google (Google Analytics 4)	Live, only with consent	Page views on public marketing pages, loaded only after the visitor accepts analytics cookies; no analytics inside the signed-in app	Pseudonymous cookie ID, page visited, device and approximate location	Google	Google Ads Data Processing Terms
Transactional email	Resend	Live	Sends account, bet, safer-gambling and complaint emails from an outbox	Email address, name, message content	[RESEND REGION]	Resend terms and DPA

Function	Supplier	Status	What they do	Data shared	Where processed	Contract / assurance
Error monitoring	Functional Software (Sentry)	Optional, live when `SENTRY_DSN` set	Receives error reports; session replay disabled	Error context (may include user ID)	EU (Germany) — `ingest.de.sentry.io`	Sentry terms and DPA
Sign-in	Google (OAuth)	Live, optional	Lets customers sign in with a Google account	Email, name (from Google)	Google	Google API terms
Live streaming	Kick	Live	Channel linking (OAuth), live status, webhooks, embedded video player for stream matches	Kick channel ID and tokens (tokens encrypted at rest)	Kick	Kick developer terms
Domain and DNS	Namecheap	Live	Registers playstake.org and serves its DNS	None	Namecheap	Namecheap terms
Source control and build checks	GitHub (Microsoft)	Live	Hosts source code; runs tests, dependency audit and malware scan on each change	Source code only; no customer data	GitHub	GitHub terms
Web fonts	Google Fonts	Live	Serves font files to browsers	Visitor IP address (to Google)	Google	Google terms. Consider self-hosting (§6).
Game developers (widget integrations)	Each developer	Live	Receive signed bet-lifecycle webhooks for their own games	Bet IDs, amounts, outcomes for their game	Developer's systems	Developer terms

5. What is not outsourced

- All gambling software: challenges and match management, the wallet and double-entry ledger, escrow, settlement, result verification, disputes and the referee console. (PlayStake runs no game logic or random number generation of its own.)
- Every compliance control: eligibility gate, deposit limits, breaks and self-exclusion, reality checks, customer interaction, AML scans, complaints handling, audit logging.
- Customer data processing: it runs only in PlayStake's own services on Railway.

6. Gaps and actions

#	Gap	Action	Needed for
1	~~Hosting region not recorded~~ Region recorded 27 October 2026 (EU West, Netherlands). Backup location still to record	Record the backup location here and in Document 10	Application
1a	Customer funds not yet segregated	Open the customer-funds account and switch the rating (Policy 19 §6); LCCP 4.1.1	Application
2	No location-at-play check	Residency is enforced (verified address in Great Britain). Still open: decide whether to add IP geolocation (for example MaxMind GeoLite2, or Cloudflare in front of the site) to block play from outside Great Britain, including Northern Ireland.	Launch
3	Planned suppliers not contracted	GAMSTOP, IBAS, eIDV and sanctions/PEP screening	Application / launch as marked

#	Gap	Action	Needed for
4	Supplier assurance not on file	Collect SOC 2 / ISO 27001 reports or equivalent from Railway, Stripe, Resend, Sentry; record DPAs	Application
5	Third-party fonts	Self-host fonts so no visitor data goes to Google	Recommended

7. Review

The Head of Compliance keeps this map current with engineering. Any new supplier, removal of a supplier, or change of hosting is reviewed before it goes live, reflected here, and assessed as a possible key event (Document 14).

Version	Date	Change
0.1	30 September 2026	First draft